

Master Thesis Proposal

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1 Background

Multimodal Large Language Models (MLLMs) have become one of the most influential technologies in recent years. Despite their impressive performance across a wide range of tasks, previous research has shown that such models often struggle with spatial reasoning, particularly when perspective changes are involved [1]. While understanding and reasoning about spatial relations between objects from different perspectives is a basic aspect of human communication and cognition.

Levinson [3] provided a widely adopted framework for spatial language, distinguishing three major frames of reference (FoRs): intrinsic, relative, and absolute. Dobnik et al. [2] demonstrated that priming can influence speakers' choice of frame of reference during communication. This thesis investigates whether similar phenomena can be observed in contemporary models.

2 Short Description and Motivation

This project aims to evaluate the ability of MLLMs to reason about spatial relations under different frames of reference (FoRs), and to investigate whether their FoR choices can be influenced through priming. We generate a small set of controlled 3D scenes (3–5 scenes) using Blender, rendered from multiple viewpoints. Each scene contains furniture objects with clear intrinsic orientations (e.g. sofa, chair, table, lamp), allowing spatial relations to be defined under two FoRs: relative (anchored to the speaker's viewpoint) and intrinsic (anchored to the object's own facing direction).

The first task examines whether MLLMs can distinguish between the relative and intrinsic frames of reference. We keep the scene constant and manipulate only the prompt. The model is asked the same spatial question three times about the same image, each time under a different FoR instruction. In the first condition, the question is grounded in the speaker-relative FoR (e.g. "From your perspective looking at the scene, what is to the left of the chair?"). In the second condition, it is grounded in the intrinsic FoR of an object (e.g. "From the sofa's own perspective, what is to the left of the chair?"). In the third condition, the model is asked to adopt a hearer's perspective by being told that

the interlocutor is standing opposite, facing the scene from 180° (e.g. "I am standing opposite you, facing you. From my perspective, what is to the left of the chair?"). Scenes are constructed such that the three conditions yield different correct answers, ensuring that a model which genuinely distinguishes between frames of reference will produce different responses across conditions. If the model gives the same answer regardless of the FoR instruction, this suggests it defaults to a single frame and fails to adapt its spatial reasoning accordingly.

The second task investigates whether FoR selection can be changed through priming. The model is presented with a scene and a four-turn vision question answering sequence. In Turn 1, a spatial description grounded in the intrinsic FoR is provided as context (e.g. There is a chair in front of the sofa), implicitly establishing the object's intrinsic orientation as the reference frame. In Turns 2 and 3, the model is asked an ambiguous spatial question whose answer differs depending on which FoR is applied. (e.g. Is there a chair to the left of the lamp? In this case, it's the hearer's/speaker's perspective). In Turn 4, the model is asked to freely describe the location of an object. The generated description is analysed to determine which FoR the model adopts, and whether this choice reflects the frame established in Turn 1 or defaults to an alternative frame such as the relative FoR.

References

- [1] Kristina Ahrens, Matthias Kerzel, Jong Hwa Lee, Cornelia Weber, and Stefan Wermter. Knowing earlier what right means to you: A comprehensive vqa dataset for grounding relative directions via multi-task learning. *arXiv preprint arXiv:2207.02624*, 2022.
- [2] Simon Dobnik, Christos Koniaris, and John D. Kelleher. Investigating the role of priming and alignment of perspective in dialogue. In *Proceedings of the 17th Workshop on the Semantics and Pragmatics of Dialogue (DialDam)*, pages 182–184, Amsterdam, The Netherlands, December 2013. University of Amsterdam.
- [3] Stephen C. Levinson. *Space in Language and Cognition: Explorations in Cognitive Diversity*. Cambridge University Press, 2003.